

CANONICAL TRANSFORMATIONS IN QUANTUM FIELD THEORY AND SOLITONS

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Received 22 December 1980

(Revised 9 March 1981)

In a sine-Gordon model we discuss a transformation which can be viewed as a quantum image of the Bäcklund transformation. The soliton is described in terms of condensation of quanta; their confinement is ascribed to the non-implementability of unitary transformations among unitarily inequivalent representations of canonical commutators and to the related topological conservation law.

1. Introduction

In many-body physics, as well as in high-energy particle physics, much attention has been devoted to the study of soliton-like solutions to non-linear evolution equations [1, 15]. Such solutions could be indeed useful in the description of “extended objects” such as dislocations in crystals, vortices in superconductivity or, in high-energy physics, monopoles in non-abelian gauge theories, bag models for elementary particles, etc. The quantum origin of these extended objects requires, however, an understanding of the soliton-like solutions in terms of basic quantum constituents.

It is known that for classical non-linear evolution equations the Bäcklund transformation is a canonical one which keeps the hamiltonian form invariant [2].

In the quantum field theory an operator has been constructed generating a canonical transformation among Heisenberg fields and leading to a coherent state representation for a soliton solution [3]. In the present work we consider a quantum sine-Gordon 1 + 1 dimensional model. The condition that the vacuum expectation value (v.e.v.) of the Heisenberg scalar field reproduces the static soliton solution is found to be satisfied when the LSZ asymptotic quantum field (constituent) undergoes a canonical transformation.

The same condition leads us to consider the tree approximation only.

It will be seen that although the transformation for the asymptotic field cannot be defined for $x_1 \rightarrow -\infty$ (or $x_1 \rightarrow +\infty$), the transformation for the Heisenberg field

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can be formally extended to the full x_1 axis. This fact expresses a well-known property characterizing the classical behaviour of the soliton, namely its interpolation between degenerate vacua. In sect. 3 we see that this is related, on the other hand, to the existence of a topological conservation law and to the stability of the soliton.

By successive canonical transformations of the LSZ asymptotic field, n -soliton solutions can be generated; they appear thus as the “quantum image” of the Bäcklund transformations. Their generator is considered in sect. 3, where the soliton solution is represented in terms of coherent state; the relation to the coherent representation of ref. [3] is also discussed.

Finally let us observe that the reconstruction of the soliton in terms of the v.e.v. of a quantum field has been presented in refs. [4–6]. There the tree approximation has also been used and a canonical transformation (“the boson transformation”) for the asymptotic field has been assumed. Canonical transformations from the free field base have been also used in ref. [18].

2. Canonical transformations and the soliton solution

We first introduce the quantum field theory framework. Let us consider the 1 + 1 dimensional quantum sine-Gordon model whose Heisenberg field equation is given by

$$\partial^2 \hat{\phi}(x) + \frac{m^3}{g} \sin g \frac{\hat{\phi}(x)}{m} = 0, \quad x \equiv (x_0, x_1), \quad \partial^2 \equiv \partial_0^2 - \partial_1^2. \quad (1)$$

$\hat{\phi}(x)$ is a Heisenberg scalar field. Eq. (1) can be written, upon expansion of the sine, as

$$(-\partial^2 - m^2) \hat{\phi}(x) = -\frac{m^3}{g} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{(2k+1)!} \left(\frac{g}{m}\right)^{2k+1} \hat{\phi}(x)^{2k+1}, \quad (2)$$

i.e., with obvious notation,

$$\Lambda(\partial) \hat{\phi}(x) = J[\hat{\phi}(x)]. \quad (3)$$

In the previous equations m is the physical mass. Let $\hat{\phi}^{\text{in}}(x)$ denote the LSZ asymptotic in-field (equivalently we could work with the out-field $\hat{\phi}^{\text{out}}(x)$):

$$\Lambda(\partial) \hat{\phi}^{\text{in}}(x) = 0. \quad (4)$$

Our quantum field theory is completely specified only when the Hilbert space on which the field operators are to be realized is given. We choose as our Hilbert space

the Fock space for the in-field $\hat{\phi}^{\text{in}}(x)$ denoted by $\{|0\rangle\}$, with the vacuum state $|0\rangle$:

$$\hat{\phi}^{\text{in}}(x)^{(+)}|0\rangle = 0. \quad (5)$$

$\hat{\phi}^{\text{in}}(x)^{(-)}$ denotes the annihilation operator of $\hat{\phi}^{\text{in}}(x)$. Also we assume that

$$\langle 0|\hat{\phi}(x)|0\rangle = 0. \quad (6)$$

The Heisenberg field operates on the states of $\{|0\rangle\}$ when expressed in terms of $\hat{\phi}^{\text{in}}(x)$. In other words a solution of the dynamics in terms of $\hat{\phi}^{\text{in}}$ is determined when the “dynamical map” [7]

$$\langle a|\hat{\phi}(x)|b\rangle = \langle a|F[\hat{\phi}^{\text{in}}(x)]|b\rangle \quad (7)$$

is found. Here F denotes a functional of normal products of $\hat{\phi}^{\text{in}}$ and is determined by the dynamics; $|a\rangle$ and $|b\rangle$ belong to $\{|0\rangle\}$. It is important to note that $|a\rangle$ and $|b\rangle$ are wave-packet states since $\{|0\rangle\}$ is the space of physical states.

Let us observe that eq. (3) leads to the Yang-Feldman equation

$$\langle a|\hat{\phi}(x)|b\rangle = \langle a|\left[\hat{\phi}^{\text{in}}(x) - i\int d^2y \Delta_{\text{R}}(x-y)\right]J[\hat{\phi}(y)]|b\rangle, \quad (8)$$

where $\Delta_{\text{R}}(x-y)$ is the retarded Green function:

$$\Lambda(\partial)\Delta_{\text{R}}(x-y) = i\delta^{(2)}(x-y). \quad (9)$$

Let us now turn our attention to the classical Euler equation framework. The static Euler equation is

$$\Lambda_s(\partial)\phi_c^{(1)}(x_1) = J[\phi_c^{(1)}(x_1)] \quad (10)$$

where $\Lambda_s(\partial) \equiv \partial_1^2 - m^2$ and J has the same form as in eq. (3). $\phi_c^{(1)}(x_1)$ is the c-number function for the single-soliton solution:

$$\phi_c^{(1)}(x) = -\frac{4m}{g} \text{tg}^{-1} e^{-m(x_1-a)}. \quad (11)$$

Let $f(x_1)$ denote a solution of

$$\Lambda_s(\partial)f(x_1) = 0; \quad (12)$$

e.g.

$$f(x_1) = \text{const} \cdot e^{-mx_1}. \quad (13)$$

We do not choose as $f(x_1)$ a superposition of e^{-mx_1} and e^{+mx_1} since it would be divergent both as $x_1 \rightarrow -\infty$ and as $x_1 \rightarrow +\infty$. When the constant in (13) is chosen to be $-4(m/g)e^{ma}$, $\phi_c^{(1)}(x_1)$ can be written as

$$\phi_c^{(1)}(x_1) = f(x_1) + \int dy_1 K(x_1 - y_1) J[\phi_c^{(1)}(y_1)] \quad (14)$$

$$= f(x_1) - \frac{g^2}{3!} \int dy_1 K(x_1 - y_1) f^3(y_1) + \dots, \quad (15)$$

with

$$\Lambda_s(\partial) K(x_1 - y_1) = \delta(x_1 - y_1). \quad (16)$$

The choice (13) and the convergence of the integrals in (15) require that

$$K(x_1 - y_1) = -\theta(y_1 - x_1) \frac{1}{m} \sinh m(x_1 - y_1). \quad (17)$$

Our assumption is now that a correspondence can be established between the quantum level and the classical level. To exhibit such a correspondence let us assume that there exists an invariant transformation B for the static Heisenberg field equation:

$$\hat{\phi}^{(1)}(x) = B\hat{\phi}(x), \quad (18)$$

$$\Lambda(\partial)\hat{\phi}^{(1)}(x) = J[\hat{\phi}^{(1)}(x)]. \quad (19)$$

We want to relate the v.e.v. of $\hat{\phi}^{(1)}$ to $\phi_c^{(1)}$. Since the Heisenberg field operates on the Fock space through the physical particle field $\hat{\phi}^{\text{in}}(x)$, we have to find a transformation β ,

$$\hat{\phi}^{\text{in}}(x) \rightarrow \hat{\phi}^{\text{in}'}(x) = \beta\hat{\phi}^{\text{in}}(x), \quad (20)$$

such that, as an effect of the non-linearity of the dynamical map eq. (7), it “dynamically rearranges” [8, 9] as

$$\langle a | B\hat{\phi}(x) | b \rangle = \langle a | F[\beta\hat{\phi}^{\text{in}}(x)] | b \rangle. \quad (21)$$

Furthermore, we require that β is a canonical transformation for $\hat{\phi}^{\text{in}}(x)$, i.e. it preserves the canonical commutation relation (c.c.r.)

$$[\hat{\phi}^{\text{in}}(x), \dot{\hat{\phi}}^{\text{in}}(y)]_{x_0=y_0} = i\delta(x_1 - y_1), \quad [\hat{\phi}^{\text{in}}(x), \hat{\phi}^{\text{in}}(y)]_{x_0=y_0} = 0, \quad \text{etc.} \quad (22)$$

and is an invariant transformation for the free field equation (4):

$$\Lambda(\partial)\hat{\phi}^{\text{in}'}(x) = 0. \quad (23)$$

The Yang-Feldman equation for $\hat{\phi}^{(1)}$ is

$$\langle 0|\hat{\phi}^{(1)}(x)|0\rangle = \langle 0|\hat{\phi}^{\text{in}'}(x)|0\rangle - i \int d^2y \Delta_{\mathbf{R}}(x-y) \langle 0|J[\hat{\phi}^{(1)}(y)]|0\rangle. \quad (24)$$

Observe now that

$$\langle 0|J[\hat{\psi}(x)]|0\rangle = J[\langle 0|\hat{\psi}(x)|0\rangle] + O(h), \quad (25)$$

where $O(h)$ denotes contribution of the order of h , the Planck constant, due to the contraction of in fields which create loop diagrams. Thus

$$\begin{aligned} \lim_{h \rightarrow 0} \langle 0|\hat{\phi}^{(1)}(x)|0\rangle &= \langle 0|\hat{\phi}^{\text{in}'}(x)|0\rangle - i \int d^2y \Delta_{\mathbf{R}}(x-y) J \left[\lim_{h \rightarrow 0} \langle 0|\hat{\phi}^{(1)}(y)|0\rangle \right] \\ &= \langle 0|\hat{\phi}^{\text{in}'}(x)|0\rangle - i \int d^2y \Delta_{\mathbf{R}}(x-y) [\langle 0|\hat{\phi}^{\text{in}'}(y)|0\rangle]^3 + \dots. \end{aligned} \quad (26)$$

If the v.e.v. of $\hat{\phi}^{\text{in}'}(x)$ is time independent, when the static case is considered, $\Delta_{\mathbf{R}}$ is replaced by $K(x_1 - y_1)$ and d^2y by $id y_1$. Therefore, we can put

$$\phi_c^{(1)}(x_1) = \lim_{h \rightarrow 0} \langle 0|\hat{\phi}^{(1)}(x)|0\rangle = \lim_{h \rightarrow 0} \langle 0|B\hat{\phi}(x)|0\rangle. \quad (27)$$

The limit $h \rightarrow 0$ suggests use of the tree approximation when explicit computation is to be carried out [4–6]*. To find β we compare eq. (15) with eq. (26) for the static case:

$$\begin{aligned} f(x_1) - \frac{g^2}{3!} \int dy_1 K(x_1 - y_1) f^3(y_1) + \dots \\ = \langle 0|\hat{\phi}^{\text{in}'}(x)|0\rangle - \frac{g^2}{3!} \int dy_1 K(x_1 - y_1) (\langle 0|\hat{\phi}^{\text{in}'}(y)|0\rangle)^3 + \dots, \end{aligned} \quad (28)$$

* It could be observed that in refs. [4–6] use of the tree expansion from the beginning is equivalent to finding just a perturbative solution of the classical equation. However it can be shown (cf. refs. [12, 13]) that quantum corrections, including renormalization effects, can be calculated from the dynamical map in the tree approximation. These problems are extensively analyzed in refs. [12–14] where interaction between quanta and static soliton are also studied.

so that

$$\langle 0 | \hat{\phi}^{\text{in}'}(x) | 0 \rangle = f(x_1). \quad (29)$$

In conclusion

$$\hat{\phi}^{\text{in}'}(x) = \beta \hat{\phi}^{\text{in}}(x) = \hat{\phi}^{\text{in}}(x) + f(x_1). \quad (30)$$

It should be observed, however, that $\hat{\phi}^{\text{in}'}(x)$ is *not* well definite as $x_1 \rightarrow -\infty$; so we restrict eq. (30) to the domain $x_1 > 0$. Nevertheless, the B transformation can be formally extended to the full x_1 axis since $\phi_c^{(1)}(x_1)$ [cf. eq. (27)] is everywhere well behaved*. This is related to the interpolating character of the soliton solution between degenerate vacua:

$${}_{2\pi}\langle 0 | \hat{\phi}(x) | 0 \rangle_{2\pi} = -2\pi \frac{m}{g} \lim_{x_1 \rightarrow -\infty} \phi_c^{(1)}(x_1) \lim_{x_1 \rightarrow +\infty} 0 = {}_0\langle 0 | \hat{\phi}(x) | 0 \rangle_0, \quad (31)$$

which expresses, equivalently, the existence of a topological conservation law (topological charge $Q = \phi_c^{(1)}(x_1 = +\infty) - \phi_c^{(1)}(x_1 = -\infty)$). $|0\rangle_0$ denotes the Fock vacuum introduced in eq. (5). The state $|0\rangle_{2\pi}$ denotes the degenerate vacuum for which

$${}_{2\pi}\langle 0 | \hat{\phi}(x) | 0 \rangle_{2\pi} = -2\pi \frac{m}{g}. \quad (32)$$

The related space $\{|0\rangle_{2\pi}\}$ is one of the infinitely many unitarily inequivalent representations of the c.c.r. allowed in our theory in the infinite volume limit ($x_1 \rightarrow \pm\infty$) [11]. Due to the unitary inequivalence, there does not exist a well defined unitary operator generating the transformation $|0\rangle_0 \rightarrow |0\rangle_{2\pi}$ or vice versa. Nevertheless, the interpolation between expectation values of $\hat{\phi}$ in $|0\rangle_0$ and $|0\rangle_{2\pi}$, as expressed in eq. (31), is characteristic of the non-trivial topology of the soliton $\phi_c^{(1)}(x)$. This is a manifestation of the classical nature of the soliton solution: it allows the interpolation between unitarily inequivalent representations. At the quantum level, such an interpolation is strictly forbidden since quantization requires a unique definite choice of the representation**.

Let us remark that when $\hat{\phi}$ is written through the dynamical map in terms of $\hat{\phi}^{\text{in}}$, B does not preserve the canonical commutators [12–14]

$$[\hat{\phi}(x), \dot{\hat{\phi}}(y)]_{x_0=y_0} = i\delta(x_1 - y_1), \quad [\hat{\phi}(x), \hat{\phi}(y)]_{x_0=y_0} = 0, \quad \text{etc.}$$

* It can be proved that if $\hat{\psi}(x) = \Psi(x, \hat{\psi}^{\text{in}}(x))$ satisfies the Heisenberg equation, then $\hat{\psi}'(x) = \Psi(x, \hat{\psi}^{\text{in}} + f)$ satisfies the same Heisenberg equation, with $f(x)$ the solution of the free field equation for field $\hat{\psi}^{\text{in}}$; in our case, cf. eq. (19). This is the content of the boson transformation theorem, see ref. [10].

** In problems involving order parameters $\alpha_i = {}_{\alpha_i}\langle 0 | \hat{\phi}(x) | 0 \rangle_{\alpha_i}$, $i = 1, 2, \dots$ where $|0\rangle_{\alpha_i}$ denote inequivalent degenerate vacua, soliton solutions can be relevant in the study of transitions between different α_i .

To reintegrate them (which may be interpreted as completeness relations) the inclusion of bound state modes and the introduction of a “quantum coordinate” (collective coordinate) for the soliton are required. When fluctuations of this coordinate are negligible with respect to the size of the soliton, it behaves as a classical object; otherwise it behaves as a quantum soliton. In ref. [14] an extensive discussion of these problems can be found.

Since the above considerations are independent of the particular form of J in eq. (3), they also apply to a scalar 1 + 1 dimensional $\lambda\phi^4$ theory.

In the case of the sine-Gordon model one can consider a n -soliton solution $\phi_c^{(n)}(x_0, x_1)$ with $n > 1$, which is characterized by n parameters [1, 15]. We have to boost $f(x_1)$ given in eq. (13):

$$x_1 \rightarrow x'_1 = \gamma(x_1 - ux_0) + \gamma u \frac{\Delta}{2m}$$

$$c = 1, \quad 0 < u < 1, \quad \gamma = (1 - u^2)^{-1/2}, \quad \frac{1}{2}\Delta = \frac{1}{u\gamma} \log u,$$

$$f(x_1) = \text{const} \cdot e^{-mx_1} \rightarrow \text{const} \cdot e^{-m\gamma x'_1 + m\gamma u x_0 - \Delta u \gamma / 2} = f(x_0, x_1) = f(x'_1). \quad (33)$$

Here we are partially using notations and conventions of ref. [15]. Of course $f(x'_1)$ solves the free field equation (4) due to its Lorentz invariance. As we did for the one-soliton case, we can write in this case

$$\phi_c^{(n)}(x'_1) = f^{(n)}(x'_1) + \int dy'_1 K(x'_1 - y'_1) J[\phi_c^{(n)}(y'_1)],$$

where $f^{(n)}(x'_1)$ is a solution of the free field equation (4) of the general form

$$f^{(n)}(x'_1) = f^{(n)}(x_0, x_1) = \sum_{i=1}^n f_i(x'_1). \quad (34)$$

f_i are of the form (33) and characterized by the velocity parameters u_i .

We assume now that a correspondence can be established between the quantum level and the classical level such that

$$\lim_{\hbar \rightarrow 0} \langle 0 | \hat{\phi}^{(n)}(x_0, x_1) | 0 \rangle = \phi_c^{(n)}(x_0, x_1), \quad (35)$$

with $\hat{\phi}^{(n)}$ solution of the Heisenberg field equation. Then we look for a canonical transformation $\beta^{(n)}$ for $\hat{\phi}^{\text{in}}$ such that

$$\langle 0 | \hat{\phi}^{(n)}(x_0, x_1) | 0 \rangle = \langle 0 | B^{(n)} \hat{\phi}(x_0, x_1) | 0 \rangle = \langle 0 | F[\beta^{(n)} \hat{\phi}^{\text{in}}(x_0, x_1)] | 0 \rangle, \quad (36)$$

and such that it is an invariant transformation for the free field equation (4). We can

repeat the procedure followed in the one-soliton case by changing the label (1) into (n) everywhere. Under our assumptions, we thus find $\beta^{(n)}$:

$$\hat{\phi}^{\text{in}} \rightarrow \hat{\phi}^{\text{in}'} = \hat{\phi}^{\text{in}} + f^{(n)} = \hat{\phi}^{\text{in}} + \sum_{i=1}^n f_i \quad (37)$$

as the transformation which induces $B^{(n)}$

$$\hat{\phi} \rightarrow \hat{\phi}^{(n)} = B^{(n)} \hat{\phi}. \quad (38)$$

Like eq. (30), eq. (37) must be restricted to a convenient x_1 domain to make $\hat{\phi}^{\text{in}'}(x)$ a well-defined field operator. We can also think of (37) as

$$\hat{\phi}^{\text{in}} \xrightarrow{B} \hat{\phi}^{\text{in}} + f_1 \xrightarrow{B} \hat{\phi}^{\text{in}} + f_1 + f_2 \rightarrow \dots, \quad (39a)$$

in such a way that

$$\hat{\phi} \xrightarrow{B} \hat{\phi}^{(1)} \xrightarrow{B} \hat{\phi}^{(2)} \rightarrow \dots. \quad (39b)$$

In this sense we refer to (38) [or to (39)] as to the “quantum image” of the Bäcklund transformation. Eq. (37) suggests that the soliton can be viewed as an extended object created by the condensation of $\hat{\phi}^{\text{in}}$ quanta (constituents) controlled by the f shift. By using $f^{(n)}$ as given in eq. (34) with $n = 2$, an explicit reconstruction of the 2-soliton case is presented in ref. [5]; there the n -soliton case, $n > 2$, is also discussed.

3. The coherent state representation and the confinement of the constituents

To study the coherent state representation, we introduce the generator of the transformation (30); it is given by

$$D^{\text{in}} = - \int_{-\infty}^{+\infty} dy_1 g(y_1) \hat{\phi}^{\text{in}}(y), \quad (40)$$

with $g(x_1) = \theta(x_1)f(x_1)$ and $\theta(x_1)$ the step function. Use of the c.c.r. shows indeed that

$$\hat{\phi}^{\text{in}'}(x) = e^{-iD^{\text{in}}} \hat{\phi}^{\text{in}}(x) e^{iD^{\text{in}}}. \quad (41)$$

Thus, eq. (29) gives

$$\langle 0 | e^{-iD^{\text{in}}} \hat{\phi}^{\text{in}}(x) e^{iD^{\text{in}}} | 0 \rangle = \langle f | \hat{\phi}^{\text{in}}(x) | f \rangle = f(x_1), \quad (42)$$

where we have put

$$|f\rangle = e^{iD^{\text{in}}} |0\rangle \quad (43)$$

and a restriction to a convenient x_1 domain must be introduced to make (41) well defined. Due to the isometry of the transformation induced by (40), it is

$$\langle f | f \rangle = 1. \quad (44)$$

The state (43) is thus a coherent state for $\hat{\phi}^{\text{in}}(x)$ [16]:

$$\hat{a}_k^{\text{in}} |f\rangle = \alpha(k) |f\rangle, \quad (45)$$

where $\hat{a}_k^{\text{in}}$ denotes the annihilation operator of $\hat{\phi}^{\text{in}}$ and the c-number function $\alpha(k)$ is related (in a convenient x_1 domain) to the $f(x_1)$ Fourier amplitude. By using the dynamical map (7) and eq. (42) we have

$$\begin{aligned} \langle f | \hat{\phi}(x) | f \rangle &= F[f(x_1)], \\ \langle f | J[F(\hat{\phi}^{\text{in}}(x))] | f \rangle &= J[F(f(x_1))]. \end{aligned} \quad (46)$$

Thus $\langle f | \hat{\phi}(x) | f \rangle$ solves the static Euler equation

$$\begin{aligned} \Lambda_s(\partial) \langle f | \hat{\phi}(x) | f \rangle &= J[\langle f | \hat{\phi}(x) | f \rangle], \\ \langle f | \hat{\phi}(x) | f \rangle &= \phi_c^{(1)}(x_1). \end{aligned} \quad (47)$$

Again we stress that, although (42) is restricted to $x_1 > 0$ only, eq. (47) can also be formally extended to $x_1 < 0$, which expresses the interpolating character of the soliton between degenerate vacua [cf. eq. (31)].

Let us note that $|f\rangle$ is not the soliton coherent state for $\hat{\phi}(x)$ as usually given in the literature (cf. refs. [3, 17, 18]), i.e.

$$|\phi_c^{(1)}\rangle = e^{iD} |0\rangle \quad (48)$$

with

$$D = - \int dy_1 \phi_c^{(1)}(y_1) \dot{\hat{\phi}}(y), \quad (49)$$

which is indeed different from D^{in} . Since $|0\rangle$ is the free field vacuum state, to compute the r.h.s. of (48), the dynamical map for $\hat{\phi}$ must be used to express D in terms of $\hat{\phi}^{\text{in}}$. In this way the non-linearity of the dynamics is taken into account. It

should be noticed, however, that in the infinite volume limit ($x_1 \rightarrow \pm \infty$), since the soliton solution goes to a constant ($\phi_c^{(1)}(x_1 = \pm \infty) = \text{const}^{(\pm)}$) a generator such as the one in eq. (49), cannot be safely defined for the transformation

$$\hat{\phi}(x) \rightarrow \hat{\phi}(x) + \text{const}^{(\pm)}, \quad \text{at } x_1 \rightarrow \pm \infty \text{ and } \text{const}^{(+)} \neq \text{const}^{(-)} \quad (50)$$

as is well known [19]. This fact expresses the stability of the soliton state against decay into a finite number of constituents $\hat{\phi}^{\text{in}}(x)$. By following ref. [18] indeed, we can evaluate the overlap between the soliton state $|\phi_c^{(1)}\rangle$ at $x_0 = \pm \infty$ and a state of a finite number of constituents. We compute (cf. ref. [18], sect. 3)

$$\langle 0 | \phi_c^{(1)} \rangle |_{x_0 = \pm \infty} = \exp \left[-\frac{1}{2} \int dk |\tilde{\phi}_c^{(1)}|^2 \omega_k \right], \quad (51)$$

where eq. (49) and the asymptotic field for $\hat{\phi}(x)$ have been used. In eq. (51) $\tilde{\phi}_c^{(1)}$ is the Fourier amplitude of $\phi_c^{(1)}(x_1)$ and $\omega_k = (m^2 + k^2)^{1/2}$. Since

$$\infty = \int |\phi_c^{(1)}|^2 dx_1 = \int |\tilde{\phi}_c^{(1)}|^2 dk < \int \omega_k |\tilde{\phi}_c^{(1)}|^2 dk,$$

the resulting transition probability is zero. We thus conclude that the confinement of the constituents of an extended object finds its origin in the fact that the transformation

$$\hat{\phi}^{\text{in (out)}} \rightarrow \hat{\phi}^{\text{in (out)'} } = \hat{\phi}^{\text{in (out)}} + \text{const}^{(\pm)} \quad (52)$$

is not unitarily implementable. To implement (52) an infinite amount of energy would be required, because it is not an invariant transformation for the free field equation (4). As in ref. [18], this can be expressed by saying that the existence of a topological conservation law [topological charge $Q = \phi_c^{(1)}(x_1 = +\infty) - \phi_c^{(1)}(x_1 = -\infty)$] corresponds to the soliton being a stable bound state of infinite constituents.

4. Conclusion

We have discussed the transformations which the Heisenberg and asymptotic fields undergo when one looks at the classical soliton from the point of view of quantum basic constituents*. We have considered the 1 + 1 dimensional sine-Gordon model.

Suppose that there exists a quantum field operator $\hat{\phi}^{(1)}(x)$ which solves the static Heisenberg equation, and whose v.e.v. in the $\hbar \rightarrow 0$ limit is given by the static soliton

* For an interesting example of condensation generating solitons see ref. [20]. See also refs. [18, 21] for solitons in terms of free fields.

$\phi_c^{(1)}(x_1)$. Suppose also that there exists a transformation B such that $\hat{\phi}^{(1)}(x) = B\hat{\phi}(x)$, with $\hat{\phi}(x)$ a solution of the Heisenberg equation with zero v.e.v.

When $\hat{\phi}(x)$ is expressed in terms of the LSZ asymptotic in-field we have that the transformation B is induced by

$$\hat{\phi}^{\text{in}}(x) \xrightarrow{\beta} \hat{\phi}^{\text{in}}(x) + f(x_1),$$

with $f(x_1) = \text{const} \cdot e^{-mx_1}$, $x_1 > 0$ solving the free field equation. It is interesting to note that although β is definite for $x_1 > 0$ only, B can be formally extended to $x_1 < 0$ also. We have observed that this is a manifestation of the soliton interpolation between degenerate vacua expectation values of $\hat{\phi}$:

$${}_{2\pi}\langle 0|\hat{\phi}(x)|0\rangle_{2\pi} = -2\pi \frac{m}{g} \xleftarrow{-\infty \leftarrow x_1} \phi_c^{(1)}(x_1) \xrightarrow{x_1 \rightarrow +\infty} 0 = {}_0\langle 0|\hat{\phi}(x)|0\rangle_0,$$

which expresses the non-trivial topology of the soliton ($\phi_c^{(1)}(x_1 \rightarrow +\infty) - \phi_c^{(1)}(x_1 = -\infty) \neq 0$). $|0\rangle_{2\pi}$ and $|0\rangle_0$ belong to representations of the free field c.c.r. which are unitarily inequivalent in the infinite volume limit ($x_1 \rightarrow \pm\infty$) [11]. Since quantization requires a unique definite choice of the representation, we have ascribed this interpolation property to the fact that the soliton is the solution to the classical Euler equation. A discussion of a coherent state representation has been presented in sect. 3 where the interpolating character of the soliton has been related to the existence of a topological conservation law and to the "confinement" of the constituents (stability of the soliton, see ref. [18]). In the infinite volume limit the "confinement" has indeed its origin in the non-implementability of transformation among unitarily inequivalent representations.

This paper is an extended version of a talk delivered at the Workshop on non-linear evolution equations and dynamical systems, Crete, July 9–23, 1980. J. Honerkamp, Y. Kodama, P. Kulish, V.G. Makhnov, G.P. McCauley are gratefully acknowledged for useful comments. Finally, it is a pleasure to thank Professor H. Umezawa for stimulating discussions.

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